

DATA

PREPROCESSING – PART I

Fabio Stella

Associate Professor

c/o Department of Informatics, Systems and Communication

University of Milano Bicocca



PREPROCESSING

The following concepts will be introduced:

✓ **AGGREGATION**

✓ **SAMPLING**

- RANDOM WITH/WITHOUT REPLACEMENT
- STRATIFIED SAMPLING

PREPROCESSING or **DATA PREPROCESSING** is a broad area and consists of a number of different **STRATEGIES AND TECHNIQUES** that are **INTERRELATED IN COMPLEX WAYS**.

PREPROCESSING is devoted to **MAKE THE DATA MORE SUITABLE FOR DATA MINING**.

PREPROCESSING topics are

- ✓ **AGGREGATION**
- ✓ **SAMPLING**
- ✓ **DIMENSIONALITY REDUCTION**
- ✓ **FEATURE SUBSET SELECTION**
- ✓ **FEATURE CREATION**
- ✓ **DISCRETIZATION AND BINARIZATION**
- ✓ **VARIABLE TRANSFORMATION**

AGGREGATION consists of **COMBINING TWO OR MORE RECORDS INTO A SINGLE OBJECT** (record).

Day Mins
265.1
161.6
243.4
299.4
166.7
223.4
218.2
157
184.5
258.6

State
KS
OH
NJ
?
OK
KS
MA
MO
KS
WV

?

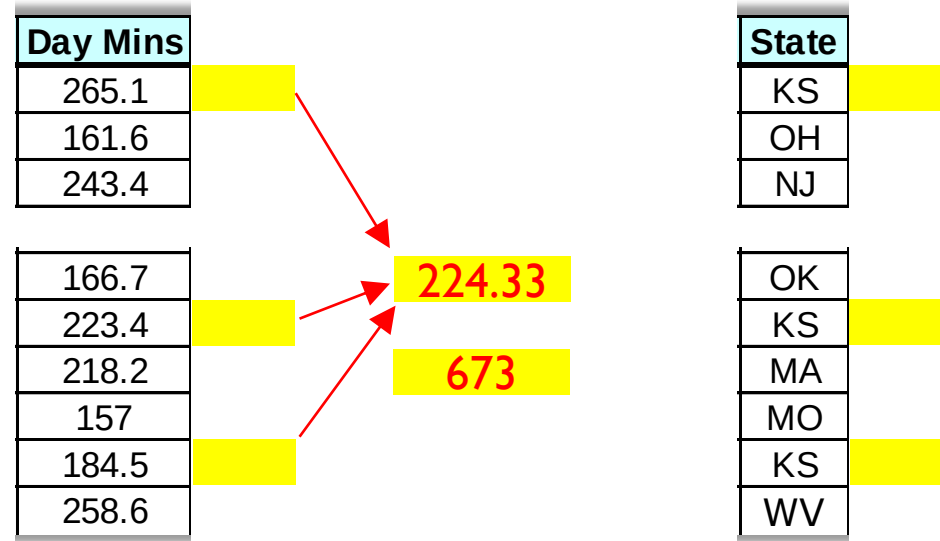
You suspect that the day calls (**DAY MINS**) depends on the state where the customer lives (**STATE**).

You want to check **HOW DAY MINS VALUES CHANGE ACROSS STATE VALUES**.

You make the decision to **AGGREGATE RECORDS ON THE BASIS OF THE VALUE OF THE STATE ATTRIBUTE**, you only **FOCUS ON STATE AND DAY MINS** attributes while **NOT PREPROCESSING THE OTHERS**.



AGGREGATION consists of **COMBINING TWO OR MORE RECORDS INTO A SINGLE OBJECT** (record).



You notice that the **4TH RECORD IS MISSING (?) THE VALUE OF THE STATE ATTRIBUTE**, you make the decision to apply the **RECORD REMOVAL STRATEGY**.

DAY MINS is a **QUANTITATIVE ATTRIBUTE** and typically such an attribute type is **MANAGED BY COMPUTING THE SUM OR THE AVERAGE** when aggregating records.

AGGREGATION consists of **COMBINING TWO OR MORE RECORDS INTO A SINGLE OBJECT** (record).

Day Mins	Churn
265.1	?
161.6	n
243.4	n
299.4	n
166.7	n
223.4	y
218.2	n
157	y
184.5	n
258.6	n

State
KS
OH
NJ
?
OK
KS
MA
MO
KS
WV



HOW TO MANAGE QUALITATIVE ATTRIBUTES SUCH AS CHURN?

AGGREGATION consists of **COMBINING TWO OR MORE RECORDS INTO A SINGLE OBJECT** (record).

Day Mins	Churn
265.1	?
161.6	n
243.4	n
299.4	n
166.7	n
223.4	y
218.2	n
157	y
184.5	n
258.6	n

State
KS
OH
NJ
?
OK
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MA
MO
KS
WV



HOW TO MANAGE QUALITATIVE ATTRIBUTES SUCH AS CHURN?

It can be **OMITTED OR SUMMARIZED** as the **SET OF ALL VALUES IT TAKES FOR ANY GIVEN VALUE OF THE AGGREGATING ATTRIBUTE**.

AGGREGATION consists of **COMBINING TWO OR MORE RECORDS INTO A SINGLE OBJECT** (record).

Day Mins	Churn		State	
265.1	?		KS	
161.6	n		OH	
243.4	n		NJ	
299.4	n		?	
166.7	n		OK	
223.4	y		KS	
218.2	n		MA	
157	y		MO	
184.5	n		KS	
258.6	n		WV	

Diagram illustrating aggregation. Two tables are shown. The first table has columns 'Day Mins' and 'Churn'. The second table has columns 'State' and 'Churn'. A yellow box labeled '{y, n}' is shown with arrows pointing to it from the 'Churn' column of the first table, indicating the aggregation of values.



HOW TO MANAGE QUALITATIVE ATTRIBUTES SUCH AS CHURN?

It can be **OMITTED OR SUMMARIZED** as the **SET OF ALL VALUES IT TAKES FOR ANY GIVEN VALUE OF THE AGGREGATING ATTRIBUTE**.

ADVANTAGES OF AGGREGATION ARE

- ✓ **SMALLER DATA SETS;** we need less memory and processing time, thus more time consuming data mining algorithms can be used.
- ✓ **CHANGE OF SCALE/SCOPE;** provides high-level view of data instead of low-level view.
- ✓ **REDUCED VARIANCE;** attributes computed on aggregated records are more stable (reduced variance for Quantitative Attributes, reduced entropy for Qualitative Attributes) than those associated with the native records. Statistical effect!!! A drawback consists of losing interesting details in the data. Aggregating sales volumes from days to weeks lost the information about which day of the week has the highest sales volume.

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	?	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	n	7	1.78	?	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	y	?	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	?	355-9993
147	0	157	y	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173

Your friend, the **TELECOM COMPANY CEO**, TELLS you that **THE CUSTOMERS CALLS DATABASE TO ANALYZE CONTAINS MORE THAN 1 BILLION RECORDS.**



You know that such a huge amount of records **DID NOT ALLOW YOU TO USE THE MOST POWERFUL** but **COMPUTATIONALLY DEMANDING DATA MINING MODELS AND ALGORITHMS**, those you hope will be the **BEST PERFORMING FOR SOLVING THE CHURN PROBLEM.**

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You make the decision to **SAMPLE A GIVEN NUMBER OF RECORDS FROM THE CUSTOMERS CALLS DATABASE**, a number of records which allows you to use computationally demanding Data Mining algorithms, those you hope will be the most successful for solving the churn problem.



A lot of **ATTENTION MUST BE PAYED TO SAMPLING**. Indeed, the **KEY PRINCIPLE FOR EFFECTIVE SAMPLING** is the following:

Using a sample will work almost as well as using the entire data set if the sample is representative.

Assume that the **MEAN OF DAY MINS** is the **PROPERTY OF INTEREST**, then a sample is representative when **DAY MINS** has approximately the same **MEAN VALUE** as the **ORIGINAL SET OF DATA (179.68)**.

MEAN VALUE 257.98



When a sample is representative?

A sample is said to be representative when it has approximately the same property (of interest) as the original set of data, i.e., the original data set.

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MEAN VALUE 257.98

MEAN VALUE 180.32

MORE REPRESENTATIVE SAMPLE



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A sample is said to be representative when it has approximately the same property (of interest) as the original set of data, i.e., the original data set.

Day Mins

161.6

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157

184.5

We must **CHOOSE A SAMPLING SCHEME THAT GUARANTEES A HIGH PROBABILITY OF GETTING A REPRESENTATIVE SAMPLE.**

This involves choosing the appropriate

- **SAMPLE SIZE**
- **SAMPLING TECHNIQUE**



Many sampling techniques exist but only the most basic ones will be presented here.

SIMPLE RANDOM SAMPLING; each record of the data set has the same probability to be included in the sample

✓ **WITHOUT REPLACEMENT;** each selected record is removed from the data set

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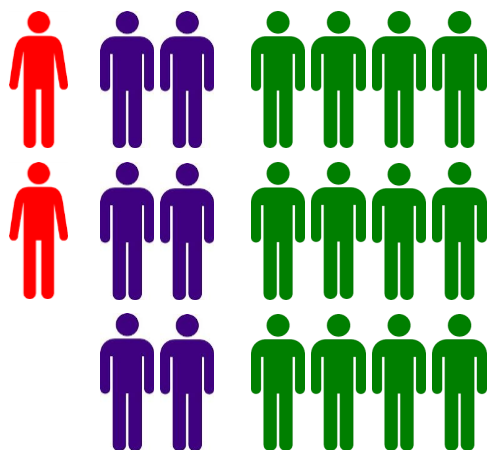
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When **SAMPLES** are **SMALL COMPARED TO DATA SET SIZE** the two sampling strategies generate **SAMPLES** which **ARE NOT MUCH DIFFERENT.**

When the data set consists of **QUALITATIVE ATTRIBUTES** such that the possible values they can take have **HIGHLY DIFFERENT FREQUENCIES**, then **RANDOM SAMPLING CAN FAIL TO ADEQUATELY REPRESENT THOSE TYPE OF VALUES WHICH ARE LESS FREQUENT.**

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DATA SET

	red	purple	green
data set	10%	30%	60%
sample	0%	50%	50%

NOT REPRESENTATIVE SAMPLE

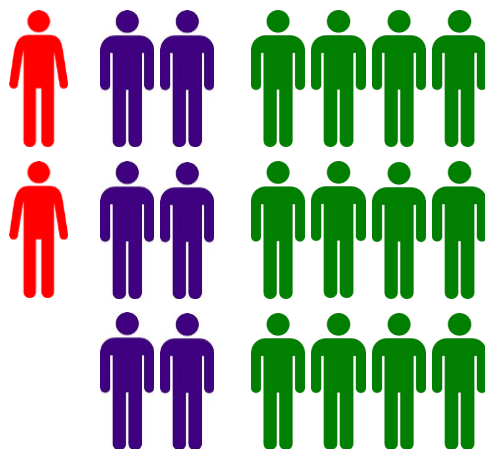


SAMPLE

Assume each **MAN'S ICON** is associated with the value of a **QUALITATIVE ATTRIBUTE** for a record of the **DATA SET** you have to analyze. Possible values are **RED**, **PURPLE** and **GREEN**.

Because **RED**, **PURPLE** and **GREEN** have **VERY DIFFERENT FREQUENCIES**, namely **2**, **6** and **12**, if we extract a sample with **RANDOM SAMPLING WITHOUT REPLACEMENT** consisting of **6 records** we **CAN OBTAIN A NON REPRESENTATIVE SAMPLE.**

When the data set consists of **QUALITATIVE ATTRIBUTES** such that the possible values they can take have **HIGHLY DIFFERENT FREQUENCIES**, then **RANDOM SAMPLING CAN FAIL TO ADEQUATELY REPRESENT THOSE TYPE OF VALUES WHICH ARE LESS FREQUENT.**



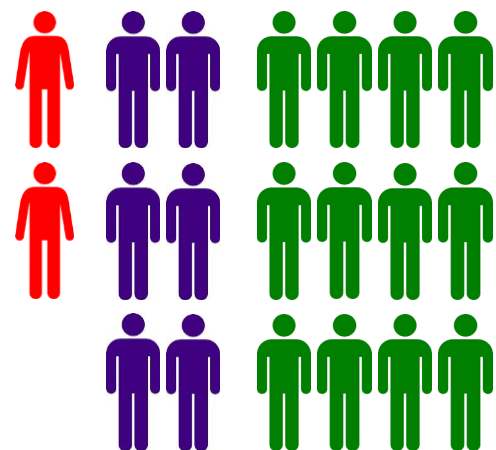
SAMPLE

DATA SET

STRATIFIED SAMPLING

EQUAL NUMBER

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DATA SET

SAMPLE

SAMPLE DIMENSION = 6

STRATIFIED SAMPLING

PROPORTIONAL

	red	purple	green
data set	10%	30%	60%
sample	0.6	1.8	3.6
rounded	1	2	3

Once a sampling technique has been selected, it is still **NECESSARY TO CHOOSE THE SAMPLE SIZE**.

- ✓ **LARGER SAMPLE SIZE INCREASE THE PROBABILITY THAT A SAMPLE WILL BE REPRESENTATIVE,** but they also **ELIMINATE MUCH OF THE ADVANTAGE OF SAMPLING**.
- ✓ **SMALLER SAMPLE SIZE, PATTERNS MAY BE MISSED OR ERRONEOUS PATTERNS CAN BE DETECTED.**

We do not give further details on **DETERMINING THE PROPER SAMPLE SIZE** which in some case can become a major issue.